HISTORY

NETWORK PROJECT

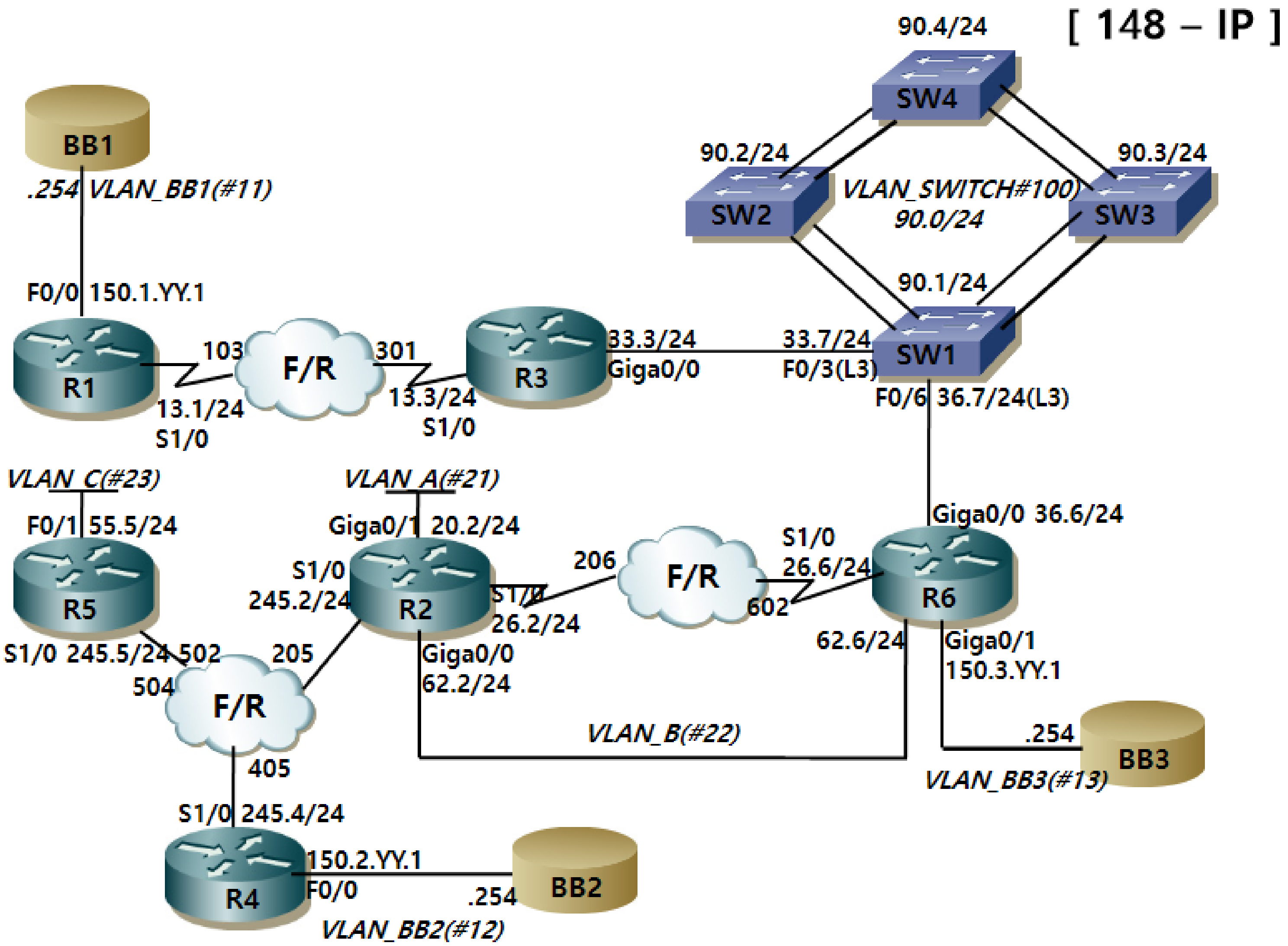
김동진 이영훈 장규혁 정성현

HISTORY

TABLE OF CONTENTS

논리적 구성도

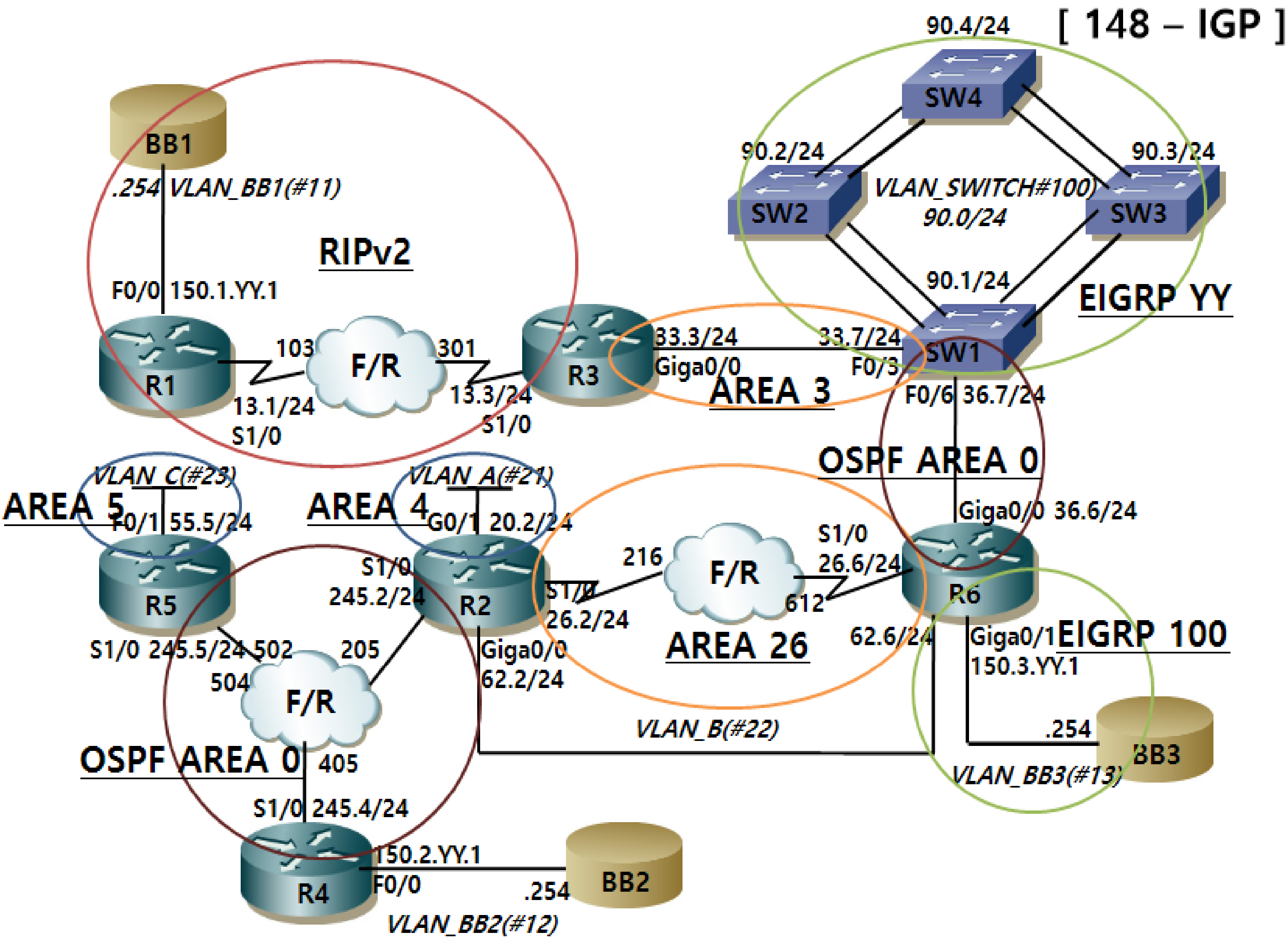
IP



[148 – IP]
90.4/24
SW4
90.2/24
SW2
VLAN_SWITCH#100)
90.0/24
90.3/24
SW3
90.1/24
SW1
BB1
.254 VLAN_BB1(#11)
F0/0 150.1.YY.1
R1
103
F/R
301
R3
33.3/24
Giga0/0
33.7/24
F0/3(L3)
13.1/24
S1/0
13.3/24
S1/0
F0/6 36.7/24(L3)
VLAN_C(#23)
VLAN_A(#21)
F0/1 55.5/24
Giga0/1 20.2/24
R5
S1/0
245.2/24
R2
206
F/R
S1/0
26.6/24
602
R6
Giga0/0 36.6/24
26.2/24
S1/0 245.5/24
502
205
504
F/R
Giga0/0
62.2/24
62.6/24
Giga0/1
150.3.YY.1
VLAN_B(#22)
.254
BB3
VLAN_BB3(#13)
405
S1/0 245.4/24
R4
150.2.YY.1
F0/0
.254
BB2
VLAN_BB2(#12)

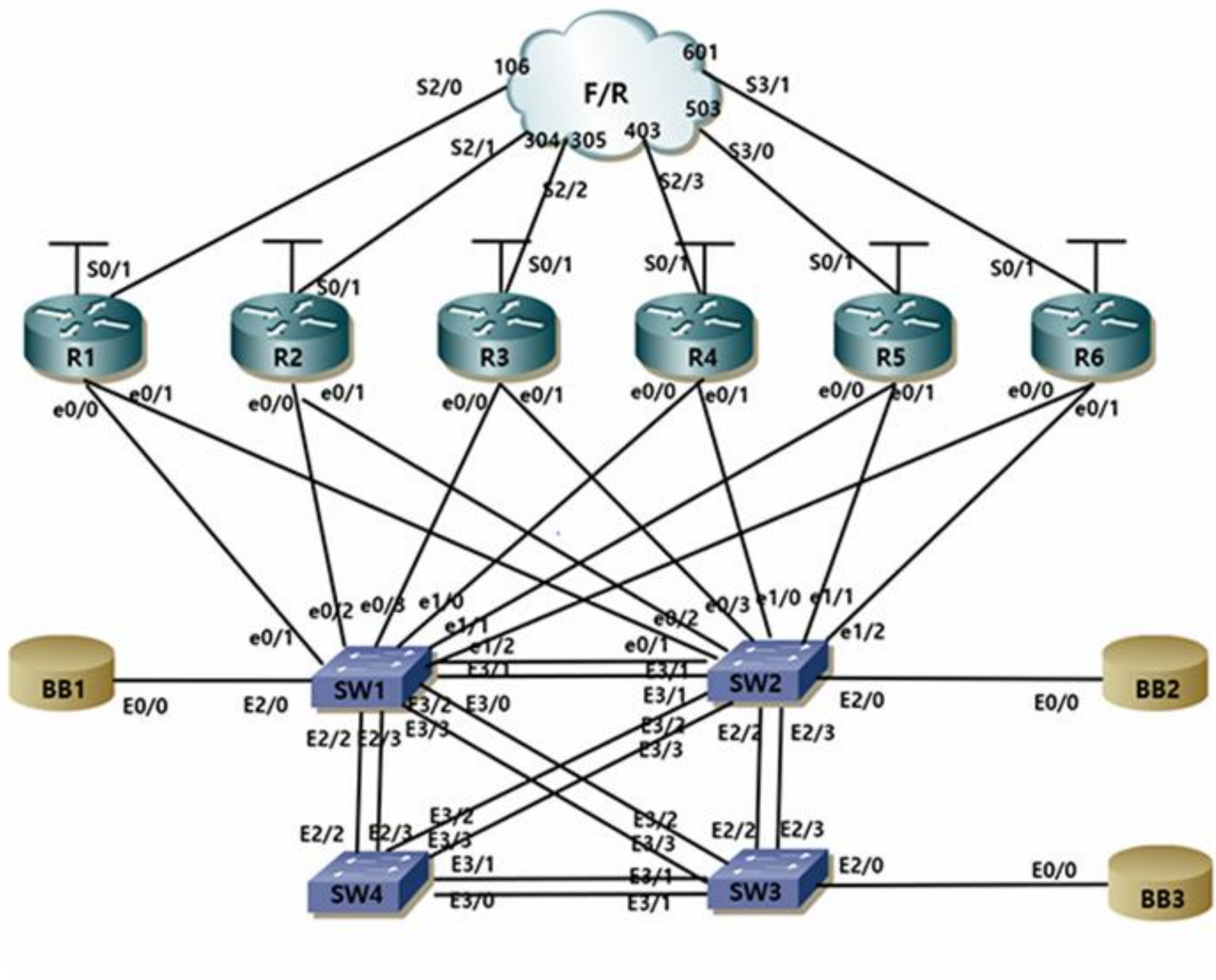
논리적 구성도

IGP



[148 – IGP]
90.4/24
SW4
90.2/24
90.3/24
SW2
VLAN_SWITCH#100)
90.0/24
SW3
90.1/24
EIGRP YY
SW1
BB1
.254 VLAN_BB1(#11)
RIPv2
F0/0 150.1.YY.1
R1
103
F/R
301
R3
33.3/24
Giga0/0
33.7/24
F0/3
13.1/24
S1/0
13.3/24
S1/0
AREA 3
F0/6 36.7/24
OSPF AREA 0
VLAN_C(#23)
AREA 5
F0/1 55.5/24
R5
VLAN_A(#21)
AREA 4
G0/1 20.2/24
S1/0
245.2/24
R2
S1/0
26.2/24
216
F/R
612
S1/0
26.6/24
Giga0/0 36.6/24
R6
S1/0 245.5/24 502
205
504
F/R
Giga0/0
62.2/24
AREA 26
62.6/24
Giga0/1
EIGRP 100
150.3.YY.1
.254
BB3
VLAN_B(#22)
VLAN_BB3(#13)
OSPF AREA 0
405
S1/0 245.4/24
R4
150.2.YY.1
F0/0
.254
BB2
VLAN_BB2(#12)

물리적 구성도



F/R
106
601
304
305
403
503
S2/0
S2/1
S2/2
S2/3
S3/0
S3/1
R1
R2
R3
R4
R5
R6
S0/1
e0/0
e0/1
SW1
SW2
SW3
SW4
BB1
BB2
BB3
E0/0
E2/0
E2/2
E2/3
E3/0
E3/1
E3/2
E3/3
e0/1
e0/2
e0/3
e1/0
e1/1
e1/2

Address Allocation

General Information

IOS 12.4

Doc CD : you have access to cisco.com/univercd

All configuration guides and master indexes are there

Tools : notepad and calculator are available

0. Address Allocation

use class B address range YY.YY.X.0/16. YY is your rack number

YY is your rack number, X is your router number

Ex (If your Rack number is 03, Lo0 is 3.3.3.3 and Rack number is 07, Lo0 is 7.7.7.7)

Note

- VLAN_A (#21) Y.Y.20.0/24
- VLAN_B (#22) Y.Y.62.0/24
- VLAN_C (#23) Y.Y.55.0/24
- VLAN_CUSTOMER1(CUSTOMER_A) (#50)
- VLAN_CUSTOMER2(CUSTOMER_B) (#79)
- VLAN_SWITCHES (#100) Y.Y.90.0/24
- VLAN_BB1 (#11)
- VLAN_BB2 (#12)
- VLAN_BB3 (#13)
- FrameRelay : YY.YY.13.0/24 (R1-R3)
  YY.YY.26.0/30(Y.Y.62.0/24) (R2-R6)
  YY.YY.245.0/24(Y.Y.24.0/24) (R2-R4-R5)
- BB1 is 150.1.YY.254/24
- BB2 is 150.2.YY.254/24
- BB3 is 150.3.YY.254/24

<loopback interface0 /24> (YY.YY.X.X : X=router number)

T/S

Frame-relay 관련 설정 미리 되어 있으며, 별도의 그림이 제공된다.

SW의 VTP mode 는 Transparent mode 로 설정 되어 있다.

SW에 vlan 설정되어 있는 부분은 없다. 문제에서 vlan 설정을 요구함.

EIGRP 구간에 있는 VLAN의 SVI interface 와 Loopback address는 설정되어 있지 않음. 문제에서 요구

R3의 G0/1 에 사용하지 않는 IP address 가 설정되어 있음(삭제)

R2,R6 serial network 와 R2,R6의 Ethernet network 가 바뀔 수 있음

Bridging and Switching

1. Configure IP across your frame relay network

frame-relay 구성 후 자기 자신의 IP address 로 ping 이 되게 설정하시오.

R1

```
int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.103 point-to-point
ip address 14.14.13.1 255.255.255.0
frame-relay interface-dlci 103
```

R3

```
int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.301 point-to-point
ip address 14.14.13.3 255.255.255.0
frame-realy interface-dlci 301
```

R2

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.245 multipoint
ip address 14.14.245.2 255.255.255.0
frame-relay map ip 14.14.245.5 205 broadcast
frame-relay map ip 14.14.245.4 205 broadcast
frame-relay map ip 14.14.245.2 205
ip ospf network broadcast

interface s1/0.206 point-to-point
ip address 14.14.26.2 255.255.255.0
frame-relay interface-dlci 206
```

R4

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

int s1/0.245 multipoint
ip add 14.14.245.4 255.255.255.0
frame-relay map ip 14.14.245.5 405 broadcast
frame-relay map ip 14.14.245.2 405 broadcast
frame-relay map ip 14.14.245.4 405
ip ospf network broadcast
```

R6

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

interface s1/0.602 point-to-point
ip address 14.14.26.6 255.255.255.0
frame-relay interface-dlci 602
```

R5

```
interface s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no shutdown

int s1/0.245 multipoint
ip add 14.14.245.5 255.255.255.0
frame-relay map ip 14.14.245.2 502 broadcast
frame-relay map ip 14.14.245.4 504 broadcast
frame-relay map ip 14.14.245.5 502
ip ospf network broadcast
```

Bridging and Switching

2.1 VTP Configuration

configure so that sw2, sw3 and sw4 will learn vlan information from sw1

SW1

vtp mode server

vtp domain history.com

vtp password history

vtp version 2

SW2

vtp mode client

vtp domain history.com

vtp password history

vtp version 2

SW3

vtp mode client

vtp domain history.com

vtp password history

vtp version 2

SW4

vtp mode client

vtp domain history.com

vtp password history

vtp version 2

SW1 결과

```
VTP Version capable              : 1 to 3
VTP version running              : 2
VTP Domain Name                  : history.com
VTP Pruning Mode                 : Disabled
VTP Traps Generation             : Disabled
Device ID                        : aabb.cc00.0700
Configuration last modified by 0.0.0.0 at 2-26-26 06:29:57
Local updater ID is 14.14.90.1 on interface Vl100 (lowest numbered VLAN interface found)

Feature VLAN:
--------------
VTP Operating Mode                 : Server
Maximum VLANs supported locally    : 1005
Number of existing VLANs           : 14
Configuration Revision             : 31
MD5 digest                         : 0x3D 0xAC 0x96 0x18 0x7F 0x0F 0x5E 0x4D
                                     0x0C 0xBC 0x85 0xB3 0x72 0xB6 0xC6 0x5C
```

2.2 Trunk Port

R6-SW2 사이를 Trunk port 로 만들고, 필요한 VLAN 정보만 흘러 다니도록 설정하라

R6

interface e0/1

no shutdown

interface e0/1.13

encapsulation dot1q 13

ip address 150.3.14.1 255.255.255.0

interface e0/1.22

encapsulation dot1q 22

ip address 14.14.62.6 255.255.255.0

SW2

interface e1/2

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk allowed vlan 13,22

Bridging and Switching

2.3 SW1,SW2,SW3,SW4

SW1-SW4

```
interface range e2/2-3
shutdown
```

SW1

```
interface range e3/0-1
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 12 mode on

interface range e3/2-3
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 13 mode on
```

SW2

```
interface range e3/0-1
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 12 mode on

interface range e3/2-3
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 24 mode on
```

SW3

```
interface range e3/0-1
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 34 mode on

interface range e3/2-3
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 13 mode on
```

SW4

```
interface range e3/0-1
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 34 mode on

interface range e3/2-3
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 24 mode on
```

SW1 결과

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------
12     Po12(SU)          -        Et3/0(P)    Et3/1(P)
13     Po13(SU)          -        Et3/2(P)    Et3/3(P)
```

2.4 Etherchannel Loadbalancing

SW1-SW2

```
port-channel load-balance src-dst-ip
```

Bridging and Switching

2.5 VLAN configuration

SW1

```
vlan 11
name BB1
vlan 12
name BB2
vlan 13
name BB3
vlan 21
name VLAN_A
vlan 22
name VLAN_B
vlan 23
name VLAN_C
vlan 50
name VLAN_CUSTOMER1
vlan 79
name VLAN_CUSTOMER2
vlan 100
name VLAN_SWITCHS

int e2/0
switchport mode access
switchport access vlan 11
int e0/0
switchport mode access
switchport access vlan 11
int e0/2
switch port mode access
switchport access vlan 22

interface e0/3
no switchport
ip address 14.14.33.7 255.255.255.0
interface e1/2
no switchport
ip address 14.14.36.7 255.255.255.0
```

SW2

```
interface e2/0
switchport mode access
switchport access vlan 12

interface e1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 13,22

interface e0/2
switchport mode access
switchport access vlan 21

int e1/1
switchport mode access
switchport access vlan 23
```

SW3

```
interface e2/0
switchport mode access
switchport access vlan 13
```

Bridging and Switching

SW1

```
SW1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/1, Et1/0, Et1/1, Et1/3
                                                Et2/1
11   BB1                              active    Et0/0, Et2/0
12   BB2                              active
13   BB3                              active
21   VLAN_A                           active
22   VLAN_B                           active    Et0/2
23   VLAN_C                           active
50   VLAN_CUSTOMER1                   active
79   VLAN_CUSTOMER2                   active
100  VLAN_SWITCH                      active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

SW2

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/0, Et0/1, Et0/3, Et1/0
                                                Et1/3, Et2/1
11   BB1                              active
12   BB2                              active    Et2/0
13   BB3                              active
21   VLAN_A                           active    Et0/2
22   VLAN_B                           active
23   VLAN_C                           active    Et1/1
50   VLAN_CUSTOMER1                   active
79   VLAN_CUSTOMER2                   active
100  VLAN_SWITCH                      active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

Bridging and Switching

SW3

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/0, Et0/1, Et0/2, Et0/3
                                                Et1/0, Et1/1, Et1/2, Et1/3
                                                Et2/1
11   BB1                              active
12   BB2                              active
13   BB3                              active    Et2/0
21   VLAN_A                           active
22   VLAN_B                           active
23   VLAN_C                           active
50   VLAN_CUSTOMER1                   active
79   VLAN_CUSTOMER2                   active
100  VLAN_SWITCH                      active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

SW4

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/0, Et0/1, Et0/2, Et0/3
                                                Et1/0, Et1/1, Et1/2, Et1/3
                                                Et2/0, Et2/1
11   BB1                              active
12   BB2                              active
13   BB3                              active
21   VLAN_A                           active
22   VLAN_B                           active
23   VLAN_C                           active
50   VLAN_CUSTOMER1                   active
79   VLAN_CUSTOMER2                   active
100  VLAN_SWITCH                      active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

Bridging and Switching

2.6 STP shared vlan

SW1	**SW4**
spanning-tree mode mst spanning-tree mst configuration name HISTORY revision 1 instance 1 vlan 11,21 instance 2 vlan 100 spanning-tree mst 1 root primary	spanning-tree mode mst spanning-tree mst configuration name HISTORY revision 1 instance 1 vlan 11,21 instance 2 vlan 100 spanning-tree mst 2 root primary
SW3	**SW2**
spanning-tree mode mst spanning-tree mst configuration name HISTORY revision 1 instance 1 vlan 11,21 instance 2 vlan 100	spanning-tree mode mst spanning-tree mst configuration name HISTORY revision 1 instance 1 vlan 11,21 instance 2 vlan 100

결과

SW1 - mst

```
SW1#show span mst conf
Name      [HISTORY]
Revision  1     Instances configured 3

Instance  Vlans mapped
--------  ----------------------------
0         1-10,12-20,22-99,101-4094
1         11,21
2         100
----------------------------------------
```

Bridging and Switching

SW1 - root

```
SW1#show span vlan 11

MST1
  Spanning tree enabled protocol mstp
  Root ID    Priority    24577
             Address     aabb.cc00.0700
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577  (priority 24576 sys-id-ext 1)
             Address     aabb.cc00.0700
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 2000000   128.1    Shr
Et2/0               Desg FWD 2000000   128.65   Shr
Po13                Desg FWD 1000000   128.514  Shr
Po12                Desg FWD 1000000   128.515  Shr
```

SW4 - root

```
SW4#show span vlan 100

MST2
  Spanning tree enabled protocol mstp
  Root ID    Priority    24578
             Address     aabb.cc00.0a00
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24578  (priority 24576 sys-id-ext 2)
             Address     aabb.cc00.0a00
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po24                Desg BLK 1000000   128.514  Shr Dispute
Po34                Desg BLK 1000000   128.515  Shr Dispute
```

Bridging and Switching

2.8 UDLD

SW3-SW4

int ran e3/0-1

udld port aggressive

2.10 Mac Address Database

SW3

mac address-table aging-time 500 vlan 13

결과

```
Global Aging Time:   300
Vlan     Aging Time
----     ----------
  13      500
```

2.11 Root Switch

SW1-SW4

spanning-tree mst configuration

instance 3 vlan 12

SW2

spanning-tree mst 3 root primary

SW1- mst

```
SW1#show span mst conf
Name      [HISTORY]
Revision  1     Instances configured 4

Instance  Vlans mapped
--------  ---------------------------
0         1-10,13-20,22-99,101-4094
1         11,21
2         100
3         12
----------------------------------------
```

Bridging and Switching

SW2 - root

```
SW2#show span vlan 12

MST3
  Spanning tree enabled protocol mstp
  Root ID    Priority    24579
             Address     aabb.cc00.0800
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24579  (priority 24576 sys-id-ext 3)
             Address     aabb.cc00.0800
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et2/0               Desg FWD 2000000   128.65   Shr
Po24                Desg LRN 1000000   128.514  Shr
Po12                Desg FWD 1000000   128.515  Shr
```

2.13 Create the interfaces VLAN below VLAN 100 (VLAN_SWITCHES)

SW1

interface vlan 100

no shutdown

ip address 14.14.90.1 255.255.255.0

SW2

interface vlan 100

no shutdown

ip address 14.14.90.2 255.255.255.0

SW3

interface vlan 100

no shutdown

ip address 14.14.90.3 255.255.255.0

SW4

interface vlan 100

no shutdown

ip address 14.14.90.4 255.255.255.0

결과

SW1

```
Interface              IP-Address      OK? Method Status                Protocol
Ethernet0/3            14.14.33.7      YES manual up                    up
Ethernet1/2            14.14.36.7      YES manual up                    up
Loopback0              14.14.7.7       YES manual up                    up
Vlan100                14.14.90.1      YES manual up                    up
```

SW2

```
Interface              IP-Address      OK? Method Status                Protocol
Loopback0              14.14.90.2      YES manual up                    up
```

SW3

```
Interface              IP-Address      OK? Method Status                Protocol
Loopback0              14.14.9.9       YES manual up                    up
Vlan100                14.14.90.3      YES manual up                    up
```

SW4

```
Interface              IP-Address      OK? Method Status                Protocol
Loopback0              14.14.10.10     YES manual up                    up
Vlan100                14.14.90.4      YES manual up                    up
```

IP IGP Protocols

1.1 RIPv2 Configuration

R1

```
router rip
version 2
no auto-summary
network 150.1.0.0
network 14.0.0.0
passive-interface default
neighbor 150.1.14.254
neighbor 14.14.13.3
```

R3

```
router rip
version 2
no auto-summary
network 14.0.0.0
passive-interface default
neighbor 14.14.13.1
```

2. EIGRP

2.1 EIGRP YY Configuration

SW1

```
ip routing
router eigrp 14
no auto-summary
network 14.14.90.1 0.0.0.0
```

SW2

```
ip routing
router eigrp 14
no auto-summary
network 14.14.8.8 0.0.0.0
network 14.14.90.2 0.0.0.0
```

SW3

```
ip routing
router eigrp 14
no auto-summary
network 14.14.9.9 0.0.0.0
network 14.14.90.3 0.0.0.0
```

SW4

```
ip routing
router eigrp 14
no auto-summary
network 14.14.10.10 0.0.0.0
network 14.14.90.4 0.0.0.0
```

2.2 EIGRP 100 configuration

R6

```
router eigrp 100
no auto-summary
network 150.3.14.1 0.0.0.0
eigrp stub redistribute
```

IP IGP Protocols

3. OSPF

3.1 AREA 0

R2

router ospf 14
net 14.14.245.2 0.0.0.0 area 0
int s1/0.245
ip ospf priority 0

R4

router ospf 14
net 14.14.245.4 0.0.0.0 area 0
net 14.14.4.4 0.0.0.0 area 0
int s1/0.245
ip ospf priority 0

R5

router ospf 14
net 14.14.245.5 0.0.0.0 area 0
neighbor 14.14.245.2
neighobr 14.14.245.4

SW1

router ospf 14
network 14.14.36.7 0.0.0.0 area 0

R6

router ospf 14
network 14.14.36.6 0.0.0.0 area 0

3.2 AREA 26 (virtual-link)

R2

router ospf 14
net 14.14.62.2 0.0.0.0 area 26
net 14.14.26.2 0.0.0.0 area 26
area 26 virtual-link 14.14.6.6

int s1/0.206
ip ospf network point-to-point
int e0/0
ip ospf network point-to-point

R6

router ospf 14
network 14.14.62.6 0.0.0.0 area 26
network 14.14.26.6 0.0.0.0 area 26
network 14.14.6.6 0.0.0.0 area 26
area 26 virtual-link 14.14.2.2

int s1/0.602
ip ospf network point-to-point
int e0/1.22
ip ospf network point-to-point

IP IGP Protocols

Ⅱ. IP IGP Protocols

3.3 AREA 3

R3

router ospf 14
network 14.14.3.3 0.0.0.0 area 3
network 14.14.33.3 0.0.0.0 area 3

SW1

router ospf 14
network 14.14.33.7 0.0.0.0 area 3
network 14.14.7.7 0.0.0.0 area 3

3.4 AREA 4

R2

router ospf 14
net 14.14.2.2 0.0.0.0 area 4
net 14.14.20.2 0.0.0.0 area 4
area 4 stub no-summary

3.5 AREA 5

R5

router ospf 14
net 14.14.55.5 0.0.0.0 area 5
net 14.14.5.5 0.0.0.0 area 5
area 5 stub

4. Redistribution

4.1 RIP – OSPF

R3

router rip
redistribute ospf 14 metric 3

router ospf 14
redistribute rip subnets

4.2 OSPF – EIGRP YY

SW1

router eigrp 14
redistribute ospf 14 metric 1 1 1 1 1

router ospf 14
redistribute eigrp 14 subnets

IP IGP Protocols

4.3 OSPF – EIGRP 100

R6

```
router ospf 14
redistribute eigrp 100 subnets

router eigrp 100
redistribute ospf 14 metric 1 1 1 1 1
eigrp stub redistribute
```

5. Loopback 어드레스는 라우팅 테이블에 /32로 보여지면 안된다.(ip ospf network point-to-point)

R1-R6

```
interface lo0
ip ospf network point-to-point
```

SW1-SW4

```
interface lo0
ip ospf network point-to-point
```

결과

Routing / Redistribute

SW1 Routing

```
      14.0.0.0/8 is variably subnetted, 18 subnets, 2 masks
O IA     14.14.2.2/32 [110/21] via 14.14.36.6, 00:02:52, Ethernet1/2
O        14.14.3.0/24 [110/11] via 14.14.33.3, 00:02:17, Ethernet0/3
O IA     14.14.6.0/24 [110/11] via 14.14.36.6, 00:02:52, Ethernet1/2
C        14.14.7.0/24 is directly connected, Loopback0
L        14.14.7.7/32 is directly connected, Loopback0
D        14.14.8.0/24 [90/130816] via 14.14.90.2, 00:02:55, Vlan100
D        14.14.9.0/24 [90/130816] via 14.14.90.3, 00:02:50, Vlan100
D        14.14.10.0/24 [90/130816] via 14.14.90.4, 00:02:37, Vlan100
O IA     14.14.20.0/24 [110/30] via 14.14.36.6, 00:02:52, Ethernet1/2
O IA     14.14.26.0/24 [110/74] via 14.14.36.6, 00:02:52, Ethernet1/2
C        14.14.33.0/24 is directly connected, Ethernet0/3
L        14.14.33.7/32 is directly connected, Ethernet0/3
C        14.14.36.0/24 is directly connected, Ethernet1/2
L        14.14.36.7/32 is directly connected, Ethernet1/2
O IA     14.14.62.0/24 [110/20] via 14.14.36.6, 00:02:52, Ethernet1/2
C        14.14.90.0/24 is directly connected, Vlan100
L        14.14.90.1/32 is directly connected, Vlan100
O        14.14.245.0/24 [110/84] via 14.14.36.6, 00:02:52, Ethernet1/2
```

IP IGP Protocols

R3 Routing

```
      14.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
R        14.14.1.0/24 [120/1] via 14.14.13.1, 00:00:14, Serial1/0.301
O IA     14.14.2.2/32 [110/31] via 14.14.33.7, 00:00:24, Ethernet0/0
C        14.14.3.0/24 is directly connected, Loopback0
L        14.14.3.3/32 is directly connected, Loopback0
O IA     14.14.6.0/24 [110/21] via 14.14.33.7, 00:00:24, Ethernet0/0
O        14.14.7.0/24 [110/11] via 14.14.33.7, 00:00:24, Ethernet0/0
C        14.14.13.0/24 is directly connected, Serial1/0.301
L        14.14.13.3/32 is directly connected, Serial1/0.301
O IA     14.14.20.0/24 [110/40] via 14.14.33.7, 00:00:24, Ethernet0/0
O IA     14.14.26.0/24 [110/84] via 14.14.33.7, 00:00:24, Ethernet0/0
C        14.14.33.0/24 is directly connected, Ethernet0/0
L        14.14.33.3/32 is directly connected, Ethernet0/0
O IA     14.14.36.0/24 [110/20] via 14.14.33.7, 00:00:24, Ethernet0/0
O IA     14.14.62.0/24 [110/30] via 14.14.33.7, 00:00:24, Ethernet0/0
O IA     14.14.245.0/24 [110/94] via 14.14.33.7, 00:00:24, Ethernet0/0
      150.1.0.0/24 is subnetted, 1 subnets
R        150.1.14.0 [120/1] via 14.14.13.1, 00:00:14, Serial1/0.301
```

R6 Routing

```
      14.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
O IA     14.14.2.2/32 [110/11] via 14.14.62.2, 00:03:13, Ethernet0/1.22
O IA     14.14.3.0/24 [110/21] via 14.14.36.7, 00:01:35, Ethernet0/0
C        14.14.6.0/24 is directly connected, Loopback0
L        14.14.6.6/32 is directly connected, Loopback0
O IA     14.14.7.0/24 [110/11] via 14.14.36.7, 00:02:10, Ethernet0/0
O IA     14.14.20.0/24 [110/20] via 14.14.62.2, 00:03:13, Ethernet0/1.22
C        14.14.26.0/24 is directly connected, Serial1/0.602
L        14.14.26.6/32 is directly connected, Serial1/0.602
O IA     14.14.33.0/24 [110/20] via 14.14.36.7, 00:02:10, Ethernet0/0
C        14.14.36.0/24 is directly connected, Ethernet0/0
L        14.14.36.6/32 is directly connected, Ethernet0/0
C        14.14.62.0/24 is directly connected, Ethernet0/1.22
L        14.14.62.6/32 is directly connected, Ethernet0/1.22
O        14.14.245.0/24 [110/74] via 14.14.62.2, 00:03:14, Ethernet0/1.22
      150.3.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        150.3.14.0/24 is directly connected, Ethernet0/1.13
L        150.3.14.1/32 is directly connected, Ethernet0/1.13
```

IP IGP Protocols

R1 Redistribute

```
     14.0.0.0/8 is variably subnetted, 21 subnets, 2 masks
C       14.14.1.0/24 is directly connected, Loopback0
L       14.14.1.1/32 is directly connected, Loopback0
R       14.14.2.2/32 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.3.0/24 [120/1] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.4.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.5.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.6.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.7.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.8.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.9.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
R       14.14.10.0/24 [120/3] via 14.14.13.3, 00:00:05, Serial1/0.103
C       14.14.13.0/24 is directly connected, Serial1/0.103
L       14.14.13.1/32 is directly connected, Serial1/0.103
R       14.14.20.0/24 [120/1] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.26.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.33.0/24 [120/1] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.36.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.55.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.62.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.90.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
R       14.14.245.0/24 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
     150.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       150.1.14.0/24 is directly connected, Ethernet0/0
L       150.1.14.1/32 is directly connected, Ethernet0/0
     150.3.0.0/24 is subnetted, 1 subnets
R       150.3.14.0 [120/3] via 14.14.13.3, 00:00:06, Serial1/0.103
```

R5 Redistribute

```
      14.0.0.0/8 is variably subnetted, 22 subnets, 2 masks
O E2     14.14.1.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA     14.14.2.2/32 [110/65] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA     14.14.3.0/24 [110/95] via 14.14.245.2, 00:01:44, Serial1/0.245
O        14.14.4.0/24 [110/65] via 14.14.245.4, 00:01:54, Serial1/0.245
C        14.14.5.0/24 is directly connected, Loopback0
L        14.14.5.5/32 is directly connected, Loopback0
O IA     14.14.6.0/24 [110/75] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA     14.14.7.0/24 [110/85] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2     14.14.8.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2     14.14.9.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2     14.14.10.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O E2     14.14.13.0/24 [110/20] via 14.14.245.2, 00:01:44, Serial1/0.245
O IA     14.14.20.0/24 [110/74] via 14.14.245.2, 00:01:45, Serial1/0.245
O IA     14.14.26.0/24 [110/128] via 14.14.245.2, 00:01:45, Serial1/0.245
O IA     14.14.33.0/24 [110/94] via 14.14.245.2, 00:01:45, Serial1/0.245
O        14.14.36.0/24 [110/84] via 14.14.245.2, 00:01:45, Serial1/0.245
C        14.14.55.0/24 is directly connected, Ethernet0/1
L        14.14.55.5/32 is directly connected, Ethernet0/1
O IA     14.14.62.0/24 [110/74] via 14.14.245.2, 00:01:45, Serial1/0.245
O E2     14.14.90.0/24 [110/20] via 14.14.245.2, 00:01:45, Serial1/0.245
C        14.14.245.0/24 is directly connected, Serial1/0.245
L        14.14.245.5/32 is directly connected, Serial1/0.245
      150.1.0.0/24 is subnetted, 1 subnets
O E2     150.1.14.0 [110/20] via 14.14.245.2, 00:01:45, Serial1/0.245
      150.3.0.0/24 is subnetted, 1 subnets
O E2     150.3.14.0 [110/20] via 14.14.245.2, 00:01:45, Serial1/0.245
```

IOS Feature / Security

2. DHCP

R6

```
ip dhcp excluded-address 14.14.62.6
ip dhcp excluded-address 14.14.62.2
ip dhcp pool DHCP
network 14.14.62.0 255.255.255.0
dns-server 150.100.1.50 150.100.1.51
domain-name history.com
lease infinite
default-router 14.14.62.6 14.14.62.2
```

3. UDP Broadcast Management

R6

```
ip forward-protocol udp bootpc
int e0/1.13
ip helper-address 150.2.14.244
```

1. OSPF Authentication

R2

```
router ospf 14
area 0 authentication message-digest
area 26 virtual-link 14.14.6.6 message-digest-key 1
md5 history
int s1/0
ip ospf message-digest-key 1 md5 history
```

R6

```
router ospf 14
area 0 authentication message-digest
area 26 virtual-link 14.14.2.2 message-digest-key 1
md5 history
interface e0/0
ip ospf message-digest-key 1 md5 history
```

R4

```
router ospf 14
area 0 authentication message-digest
int s1/0
ip ospf message-digest-key 1 md5 history
int lo0
ip ospf message-digest-key 1 md5 history
```

R5

```
router ospf 14
area 0 authentication message-digest
int s1/0
ip ospf message-digest-key 1 md5 history
```

SW1

```
router ospf 14
area 0 authentication message-digest
int e1/2
ip ospf message-digest-key 1 md5 history
```

HISTORY